

c) Solução Geral

$$y = y_h + y_p \quad (\Rightarrow) \quad y = C_1 \cos(2n) + C_2 \sin(2n) + \frac{1}{3} \sin(n), \quad C_1, C_2 \in \mathbb{R}$$

Escolha Multipln

$$\textcircled{1} \int_1^{+\infty} \frac{n}{\sqrt{n^4+1}} dn \quad e \quad \int_1^{+\infty} \frac{3}{\sqrt{n}} dn$$

$$3 \int_1^{+\infty} \frac{1}{n^{1/2}} \quad (\alpha = \frac{1}{2} \leq 1 \rightarrow \text{div})$$

$$g(n) = \frac{n}{n^2} = \frac{1}{n} \leftarrow \text{div} (\alpha \leq 1 \rightarrow \text{div})$$

$$L = \lim_{n \rightarrow +\infty} \frac{\frac{n}{\sqrt{n^4+1}}}{\frac{1}{n}} = \lim_{n \rightarrow +\infty} \frac{n^2}{\sqrt{n^4+1}} = \dots = 1 \therefore \text{Pelo C.Limite é div.}$$

⑥ Prove que $y = y_h + y_p$ é solução de uma EDO linear de ordem n

$$a_0 (y_h + y_p)^{(n)} + a_1 (y_h + y_p)^{(n-1)} + \dots + a_{n-1} (y_h + y_p)' + a_n (y_h + y_p) = b(n)$$

$$(\Rightarrow) a_0 y_h^{(n)} + a_1 y_h^{(n-1)} + \dots + a_{n-1} y_h' + a_n y_h + a_0 y_p^{(n)} + a_1 y_p^{(n-1)} + \dots + a_{n-1} y_p' + a_n y_p = b(n) (\Rightarrow)$$

$$(\Rightarrow) 0 + b(n) = b(n) (\Rightarrow) b(n) = b(n) \text{ p.v}$$